This is of the form $\int e^x[f(x)+f'(x)]dx$

$$\int e^x\left(\frac{1}{x}-\frac{1}{x^2}\right)dx = e^x\frac{1}{x}+c.$$

(ii) Let $I = \int e^x(\sin x+\cos x)\,dx$

Take $f(x) = \sin x$, then $f'(x)=\cos x$

This is of the form $\int e^x[f(x)+f'(x)]\,dx$

$$\int e^x(\sin x+\cos x)\,dx = e^x\sin x+c.$$

(iii) $\int e^x\left(\frac{1-x}{1+x^2}\right)^2dx$

$$\text{Let } I = \int e^x\frac{(1-x)^2}{(1+x^2)^2}dx$$

$$= \int e^x\frac{(1+x^2-2x)}{(1+x^2)^2}dx$$

$$= \int e^x\left(\frac{1}{(1+x^2)}-\frac{2x}{(1+x^2)^2}\right)dx$$

$$\text{If } f(x) = \frac{1}{(1+x^2)}, \text{ then } f'(x)=-\frac{2x}{(1+x^2)^2}$$

Using $\int e^x\left(f(x)+f'(x)\right)dx = e^xf(x)+c$

$$\int e^x\left(\frac{1-x}{1+x^2}\right)^2dx = \int e^x\left(\frac{1}{(1+x^2)}-\frac{2x}{(1+x^2)^2}\right)dx = e^x\frac{1}{(1+x^2)}+c.$$

EXERCISE 11.9

Integrate the following with respect to x:

(1) $e^x(\tan x+\log\sec x)$ (2) $e^x\left(\frac{x-1}{2x^2}\right)$ (3) $e^x\sec x(1+\tan x)$

(4) $e^x\left(\frac{2+\sin 2x}{1+\cos 2x}\right)$ (5) $e^{\tan^{-1}x}\left(\frac{1+x+x^2}{1+x^2}\right)$ (6) $\frac{\log x}{(1+\log x)^2}$

11.7.9 Integration of Rational Algebraic Functions

In this section we are going to discuss how to integrate the rational algebraic functions whose numerator and denominator contains some positive integral powers of x with constant coefficients.

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